



Elliptic - Predicting Illicit Transactions in Cryptocurrency



le wagon

FraudBusters

Meet the team



Kim Ean
Koh



Thomas
Ayrault



Jakob
Holmgren



Neha
Mohan



FraudBusters



le wagon

The Problem: Illicit Activity at Scale

SCALE

~\$750B

Estimated illicit crypto funds laundered through Europe each year.

ENFORCEMENT

\$4.3B

Binance was fined for failure to report 100K suspicious transactions

ENFORCEMENT

\$1.5B

CryptoMixer was taken down by Europol for laundering since 2016.

Cryptocurrency transactions are increasingly used for both legitimate and illegitimate purposes.



le wagon

The need: fraud to start with

WHY DETECT EARLY

Reduce harm and stay compliant

- \$ Protect society: prevent **fraud**, money laundering, disrupt terrorism financing, ransomware, trafficking
- \$ Regulatory compliance: meet AML/KYC obligations, avoid **finances**
- \$ Reduce enterprise risk: limit reputational and operational **losses**

THE OUTCOME WE TARGET

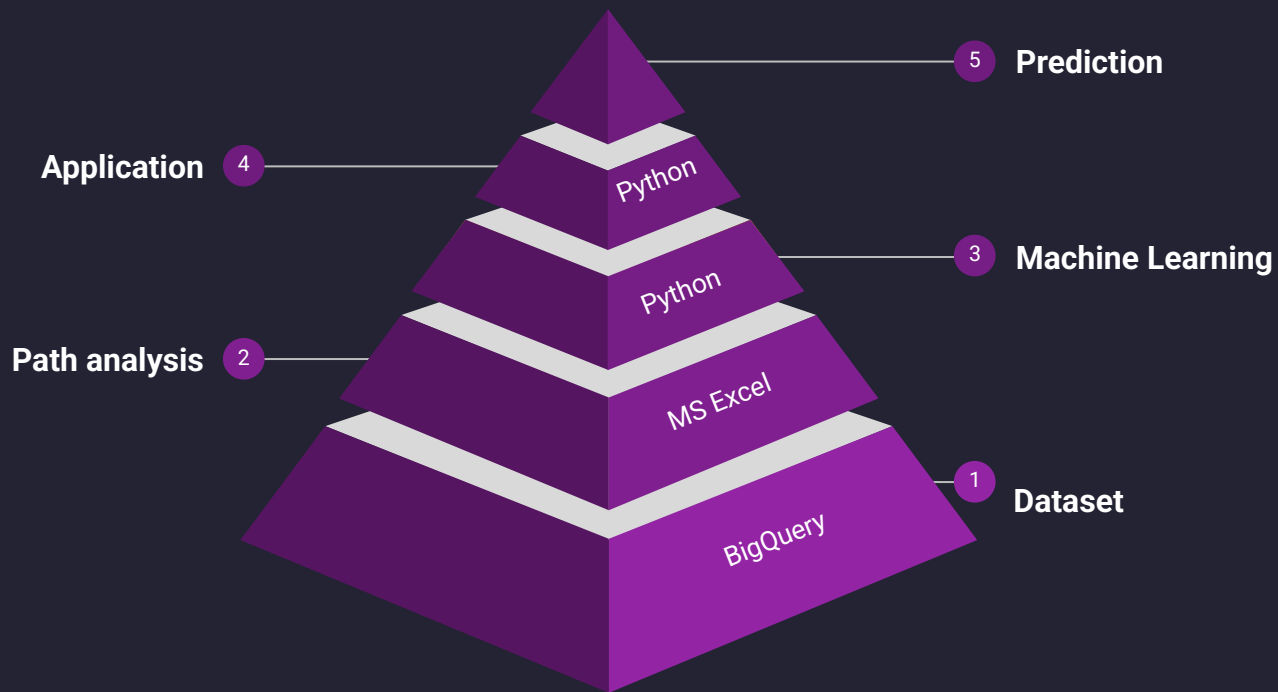
Reduce harm and stay compliant

- \$ Exploring patterns in Bitcoin transactions to distinguish licit from illicit activity
- \$ Build a predictive model to identify suspicious transactions



le wagon

Our Approach at a Glance



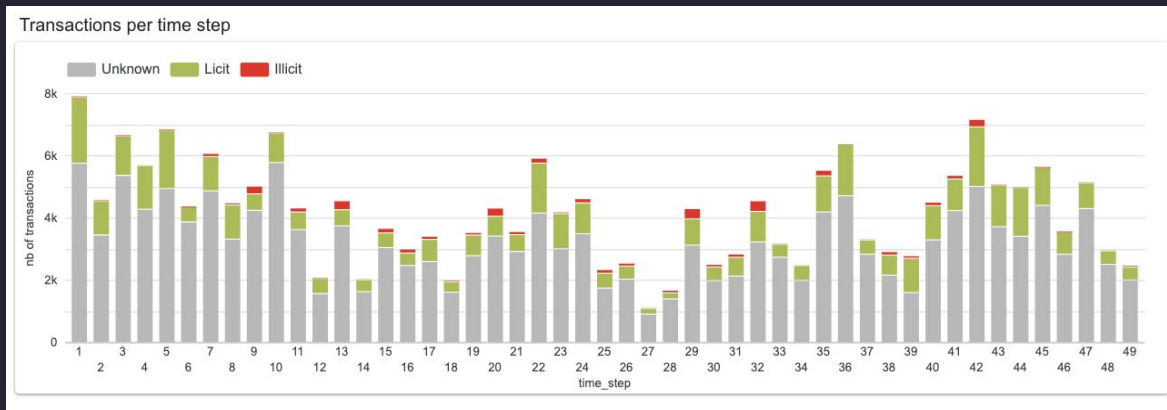
le wagon

Step 1: Initial Baseline

ELLIPTIC

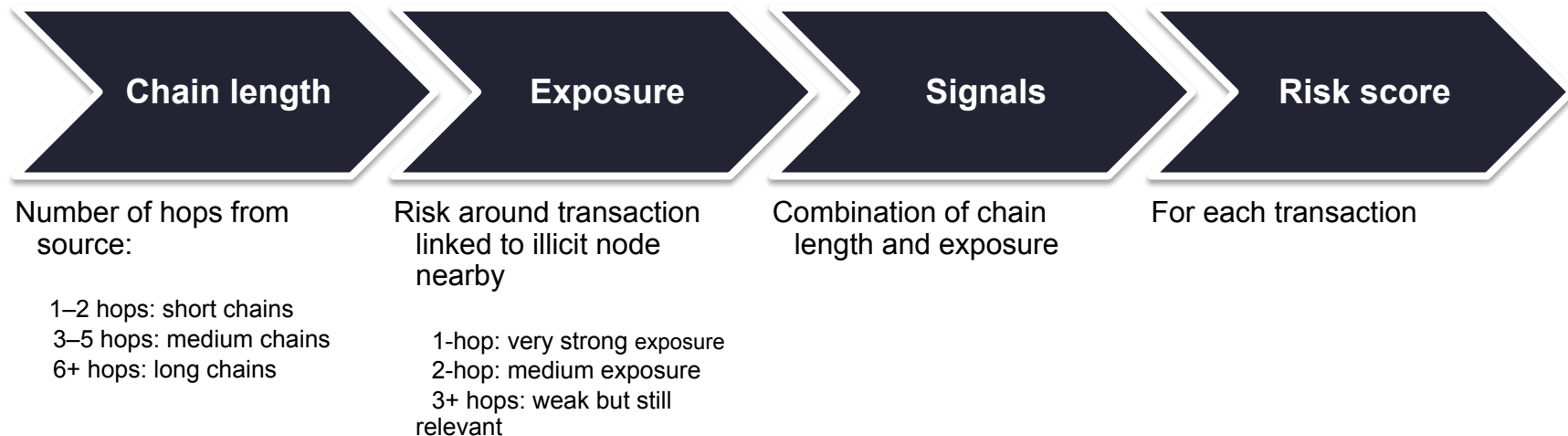
Fintech company, specializing in the use of **blockchain analytics** for **financial crime compliance**

DATASET



le wagon

Step 2: Path Analysis



Step 2: Tiering conclusion

Risk tier	Meaning	Financial Action
Tier 1 – Deep & Close	Critical laundering intermediaries	High-confidence laundering candidates Deep transaction chains and close to illicit activity > Immediate action: blocking, blacklisting, etc
Tier 2 – Deep Only	Structurally complex (long chains) but no known illicit link	Large structural-risk population Deep chains, currently no illicit proximity Exchanges hubs potential future laundering paths > Monitor candidates
Tier 3 – Close Only	Near illicit activity but not deeply layered	Directly exposed to illicit activity Shallow chains Often immediate neighbors > Early warning
Tier 4 – Low Risk	Background traffic with shallow chains and no illicit proximity	> Low operational priority



le wagon

Step 3: Machine Learning - XGBoost Performance

PERFORMANCE SCORE WITH XGBOOST

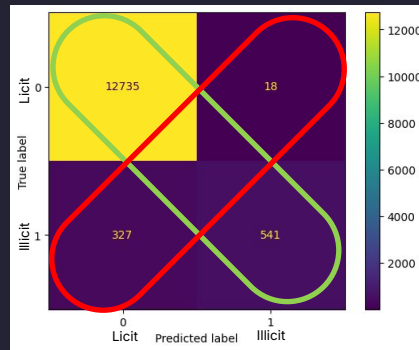
Class	Precision	Recall	F1-score
Illicit	98%	60%	75%
Licit	97%	100%	99%

Rows: 46K

Features: 166

Model: XGBoost (supervised)

CONFUSION MATRIX



True positive / negative ✓

False positive / negative X

High precision reduces false positives (18)

Trade-off between recall & precision



le wagon

Step 4: Application



le wagon



le wagon

Thank you for your time!

Appendix



le wagon

Thank you very much!



Kim Ean
Koh



Thomas
Ayrault



Jakob
Holmgren



Neha
Mohan



le wagon

Intro: 30s Thomas

Context: 30s Thomas

Approach: 30s Jakob

Path analysis: 1min Thomas

ML: 30s Jakob

App: 1.5-2min Kim



le wagon

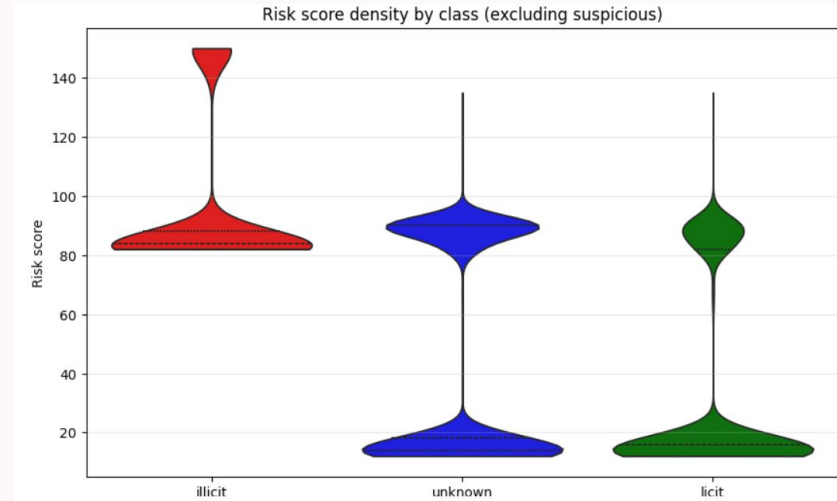
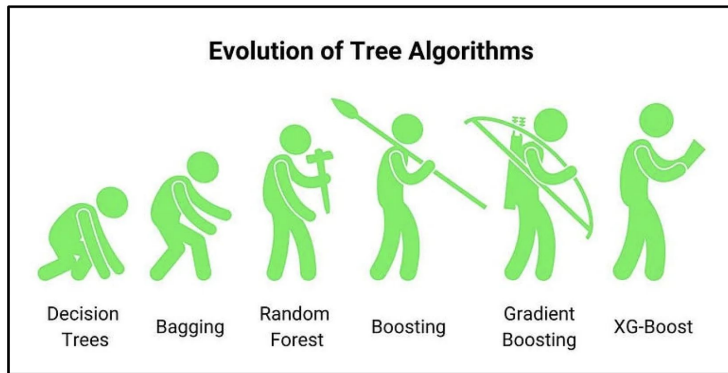
Sources

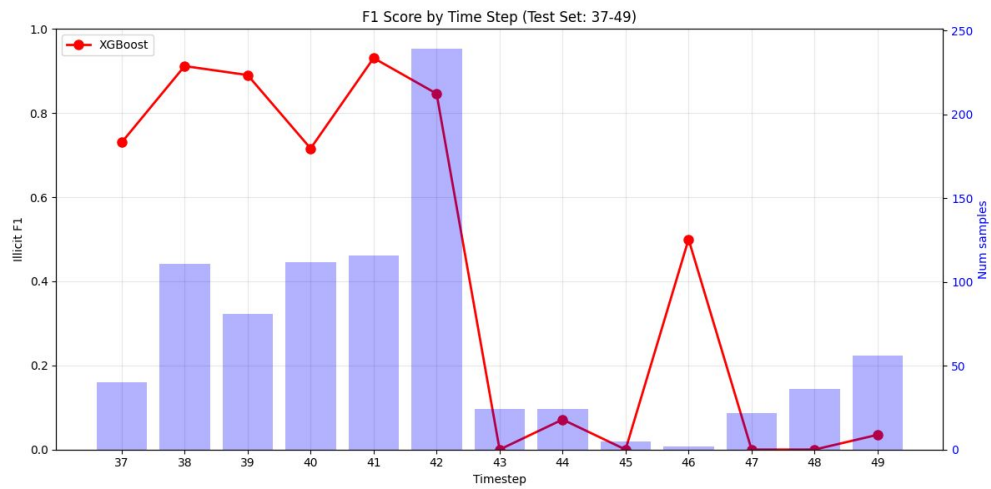
1. <https://coinlaw.io/money-laundering-statistics/>
2. <https://www.nbcnews.com/tech/tech-news/binance-ceo-steps-crypto-platform-hit-4-billion-fines-rcna126071>
3. <https://www.coindesk.com/policy/2025/12/01/european-authorities-seize-usd1-51b-bitcoin-laundering-service-cryptomixer>



le wagon

<https://www.kaggle.com/datasets/ellipticco/elliptic-data-set/data>





Row	deanonimized_id	txId	time_step	class_label	class_std	class_binary	double_field_2	total_out_value	fee	double_field_5	inputs_count	outputs_count	inputs_unique_co...
1	fc7971538ad505c0457c12a39ce580b80bced86c1cf2b1126f246659cc24876b	94769285	7	2	licit	non-illicit	0.008034489162...	-0.075894176882...	-1.185820765608...	0.253519457394...	-0.068726225245...	0.293342437339...	-0.101008611429...
2	f73e76dc9d3456f866eecef4ef7c196c40d5dd5ad6f1965306dd6d21d4cab2af	70859417	42	unknown	unknown	non-illicit	-0.055386908348...	1.443410211160...	2.064343844082...	3.461193185673...	-0.068726225245...	-0.167504573178...	15.64349968067...
3	f6f9704ef536486c550e6f32a9a639d9655ed7517bde12ebbfec2c7042df47229	179084664	9	unknown	unknown	non-illicit	-0.054959805495...	-0.047415324976...	1.522649742467...	1.353293307090...	-0.068726225245...	1.592093103343...	-0.101008611429...
4	f6b09392950a611d508a27ed9ddb78bb5669378e60f983d0aa2960242c73763b	230658142	1	unknown	unknown	non-illicit	-0.050077840218...	0.016270320709...	0.980955640851...	4.341012265430...	-0.068726225245...	5.006550499450...	-0.101008611429...

Path Analysis

1. **Find the chain length:** the maximum number of hops you can go downstream from source.
 - Short chains (1–2 hops)
 - Medium chains (3–5 hops)
 - Long chains (6+ hops)
2. **Find the exposure** or the risk of the transaction to be illicit because it is close to an illicit node using k-hop exposure
 - 1-hop → very strong exposure
 - 2-hop → medium exposure
 - 3+ hops → weak but still relevant
 -
3. Look at both the signals together is the transaction is deep and near illicit
4. Give a risk score for each transaction



Risk Scoring

Methodology

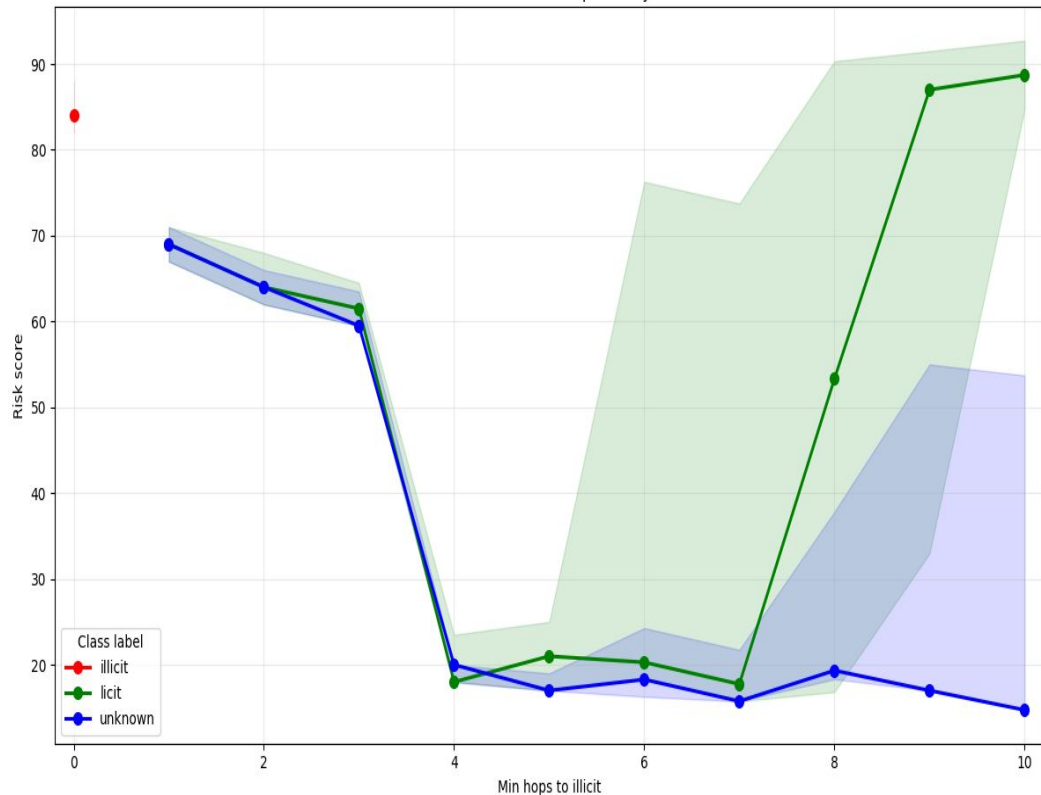
Risk tier	Meaning	Base score
Tier 1 – Deep & Close	Critical	100
Tier 2 – Deep Only	High	70
Tier 3 – Close Only	Medium	50
Tier 4 – Low Risk	Low	10

$$\begin{aligned}\text{Risk score} = & \text{Base_score} \\ & + 20\% * (\text{chain contribution}) \\ & + 30\% * (\text{exposure contribution})\end{aligned}$$



le wagon

Risk score vs illicit proximity



FraudBusters



